

Workout Research - Squats

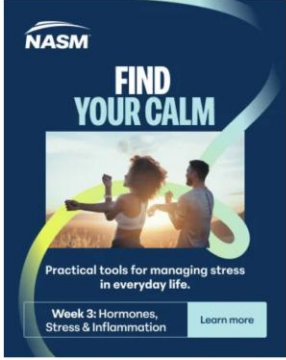
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CES - EXERCISE SCIENCE

THE MUSCLES USED IN SQUATS - SQUAT BIOMECHANICS EXPLAINED





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The squat is arguably the most popular exercise used by athletes and fitness enthusiasts alike, and for good reason. There is ample evidence describing its use for improving lower body muscular endurance, strength, muscle size, and power.

This article discusses the traditional barbell back squat from a fitness perspective. It does not discuss the squat as it relates to performance such as competing in powerlifting or Olympic Weightlifting.

Rather, the purpose of this article is to provide an overview of the movement requirements, involved musculature, common technique errors, and recommendations to maximize safety and performance.

A lot of these principles are covered in the [Corrective Exercise Course](#).

Learning Objectives:

1. Comprehend the movement requirements, joint actions, and involved musculature of the squat exercise.
2. Identify common faulty movement patterns during the squat exercise.
3. Provide exercise technique recommendations for fitness enthusiasts wishing to perform the squat exercise.

INTRODUCTION

The squat is one of the most debated exercises in the fitness and sports community, but it is hard to argue its effectiveness. There is ample evidence describing its use for improving lower body muscular endurance, strength and power (Clark, Lambert & Hunter, 2012; Folland & Williams, 2007; Marques et al., 2015; Soriano, Jiménez-Reyes, Rhea, & Marín, 2015). There are several variations of the squat exercise including the bodyweight squat, barbell back squat, barbell front squat, dumbbell squat, sumo squat, [split squat](#), box squat, plie squat, squat jump, overhead squat, and single-leg squat, to name a few.

For the sake of this article we will discuss the barbell back squat from a fitness perspective. We will not discuss the squat as it relates to performance, such as powerlifting. Instead, we will discuss the safest variation of the squat exercise for a fitness enthusiast seeking to improve technique and minimize faulty movement patterns and potential injury.

It is important to note variations of the squat exercise exist to maximize 1 repetition maximum (1RM) potential, such as using an excessively wide stance with a toe out posture. This posture reduces the amount of **hip** and **knee flexion** and **ankle dorsiflexion** needed to reach full depth. While from a biomechanical perspective this variation enables the lifter to complete the exercise with higher loads because range of motion is reduced, it may not be the safest variation on articulating joint surfaces for beginning exercisers who have no desire for improving their 1RM. Experienced individuals or athletes seeking to improve 1RM can use these variations once properly instructed and have displayed adequate physical capabilities.

Hip Flexion: Decreasing the angle between the femur (thigh) and pelvis. This occurs from a standing position when a person elevates their knee toward their abdomen (femoral-on-pelvic hip rotation) or when bending forward from the trunk, as if touching their toes (pelvic-on-femoral rotation).

Knee Flexion: Decreasing the angle between the lower leg (tibia, fibula) and femur. This occurs when a person bends their knee, bringing their heel closer to their thigh or butt.

Ankle Dorsiflexion: Flexion at the ankle in which the top of the foot (dorsal) is brought closer towards the shin.

TECHNIQUE

Starting Position

> Stand with feet approximately shoulder-width apart, toes pointing straight ahead, and knees aligned over second and third toes.

- As much as 5-8° of external foot rotation is allowed in the starting position as some consider this normal anatomical position (Schoenfeld, 2010).

> To perform the high-bar back squat, rest the barbell on the shoulders, behind the neck, with hands grasping the bar wider than shoulder-width apart.

> To perform the low-bar back squat, rest the barbell on the middle trapezius region with hands grasping the bar wider than shoulder-width apart.

- It is important to note adequate shoulder mobility (external rotation) is required to hold the bar securely.

Movement Pattern

> Slowly begin to squat down by hinging at the hips and then flexing at the knees.

> Allow glutes to "stick" out behind the body as if sitting into a chair.

> Keep the chest up and the cervical spine in a neutral position. Avoid excessive cervical flexion, extension, or anterior translation (jutting the head forward).

> Squat to a depth that can be safely controlled with no movement compensations.

- Common movement compensations include knee valgus (knock knees), rounding or arching of the low-back, an excessive forward lean of the torso, and overly externally rotating or pronating the feet.

WHAT IS THE RIGHT SQUAT DEPTH?

A blanket statement regarding squat depth for all individuals is inappropriate at best. Depending on the person's physical capabilities and goals, some should squat to a depth in which the thighs are always parallel to the floor. Others can do full squats (below parallel). Conversely, some experts recommend a limited range of motion (i.e., ¼ squat) to avoid stress on an individual's knees.

Due to individual differences in shape, size, and overall fitness, people inherently display differences in joint mobility, joint stability, and neuromuscular control (coordinated muscle activation). Individuals who display ample mobility and joint range of motion, combined with optimal joint stability, may be able to safely perform squats using a full or near full range of motion.

This typically requires at least 15-20° of ankle dorsiflexion and 120° of hip flexion (Greene, 1994). Research suggests if an individual possesses less than adequate ankle dorsiflexion, they may be at greater risk of injury to the knees, hips, or low-back during functional movement patterns (Lun, Meeuwisse, Stergiou, & Stefanyshyn, 2004; Powers, 2003). In other words, each joint must exhibit proper range of motion for the efficient transference of forces throughout the body to produce ideal movement.

According to Schoenfeld (2010), individuals with a history of patellofemoral injury should limit the depth of their squat. This recommendation is due to the fact that peak compressive forces at the knee occur at near maximum knee flexion angles. In addition, those with existing knee injury or previous reconstruction of the posterior collateral ligament (PCL- ligament on the back of the knee) should restrict knee flexion to 50-60° to minimize posterior shear forces. Moreover, muscular development of the quadriceps is maximized while performing squats to a depth with thighs parallel to the floor.

There appears to be no benefit to quadriceps development if a person performs squats to a full depth (below parallel). Yet, Schoenfeld explains, hip development is maximized when performing below parallel squats and may be important for individuals needing to perform this movement pattern (such as powerlifters or Olympic weightlifters). However, for a fitness client seeking to improve general fitness, below parallel squats are not recommended until adequate levels of stability and mobility are attained.

IDENTIFYING IDEAL SQUAT DEPTH

An easy test can be performed to identify ideal squat depth. A person should perform a barefoot squat using a mirror or a partner to evaluate his or her mechanics. During the eccentric phase of the squat, an individual's torso and shin angle should remain parallel (see Figure 1). In addition, there should be no excessive arching or rounding of the low back. Lastly, look for any faulty movement patterns at the foot/ankle. The feet should not excessively pronate (arches collapse) or externally rotate during the eccentric phase.

Once any of these movement compensations have been observed, the squat is at a depth no longer suited for the individual. He or she will need to stop just before any these faulty movements occur. By maintaining ideal posture and technique throughout the movement, he or she will develop ideal motor skills needed for this exercise. As mobility and stability improve, the individual will be able to successfully squat to deeper depths.

FEET TURN OUT EXCESSIVELY / FEET FLATTEN / HEELS RAISE

If an individual lacks adequate mobility of the ankle complex (limited ankle dorsiflexion), he or she will likely gain additional range of motion by altering foot mechanics. This usually comes in the form of excessively turning the feet outward, pronation at the foot/ankle complex, or raising the heels off the floor.

When someone lacks ankle dorsiflexion, which occurs in the sagittal plane, the range of motion must then take place in another plane (frontal or transverse). Excessive external rotation of the feet (beyond 8°) enables a person to squat to a lower depth because motion is occurring primarily in the transverse plane (Figure 2). In other words, due to limited ankle mobility, the knees are not able to track over the toes in the sagittal plane, so motion is borrowed from another plane. This may be caused by tightness in the calf complex (gastrocnemius, soleus) and/or restriction in the talocrural (ankle) joint.

Some research indicates restriction in ankle mobility may cause knee valgus (knock knees), which is often a recipe for patellofemoral pain or even ACL injury (Bell, Oates, Clark, & Padua, 2013; Dill, Begalle, Frank, Zinder, & Padua, 2014; Macrum, Bell, Boling, Lewek, & Padua, 2012). In these instances, flexibility exercises for the calves and possibly joint mobilization for the ankle may be required to regain 15-20° of ankle dorsiflexion.

Sagittal Plane: An imaginary plane that bisects the body into right and left sides. Movements in the sagittal plane include flexion and extension, such as knee flexion/extension, hip extension/flexion and shoulder extension/flexion.

Frontal Plane: An imaginary plane that bisects the body into front and back halves. Movements in the frontal plane include abduction and adduction, such as hip adduction/abduction and lateral trunk flexion (side bending).

Transverse Plane: An imaginary plane that bisects the body into top and bottom halves. Movements in the transverse plane include rotational movements, such as trunk rotation, hip internal/external rotation and shoulder internal/external rotation.

Workout Research – Biceps Curls

[Link to the website](#)

Beginner Biceps Workout

My first memory of wanting to build an impressive physique was as a 15 year old looking at the cover of Men's Health, and wanting to have the same peaked biceps and sculpted six pack abs. I was fortunate to have some guidance from an early age and so was quick to learn the basic movements and establish proper form and technique, which helped build a strong foundation for where I am today.

Within this article, I will focus on covering three different bicep-focused routines: Beginner, Intermediate, and Advanced, which each section accompanied by it's own video.

It's not so much that the beginner routine is only for people who have little to no experience training, and the advanced is only for, say competitor and professional athletes with many years of experience. It's simply based around fundamentals and when and how to employ certain shocking principles. Still to this day, I will follow many of the same fundamental basics as shown within the beginner routine, yet mix it up with many of the techniques and methods discussed throughout the intermediate and advanced routines. I would encourage you to do the same.

The Warm Up

As when training any muscle group, I would strongly recommend performing a few warm up exercises, along with some specific stretching and mobility. The main benefit to this is that it helps to pool blood (along with oxygen and nutrients into the working muscles), and helps to properly prepare the joints and tendons for the stresses ahead. I also find this beneficial to our psyche, and helps us to mentally prepare for the lifts ahead.

I've shown within the first section of the beginner routine, a great bicep warm that consists of banded chin ups, rotational dumbbell curls, inverted plank, and straight arm, behind-the-back, bicep stretch. The emphasis here is not on weight (in fact, you shouldn't be getting anywhere close to muscle failure), but rather working the muscles through their full range of motion across a variety of different angles and positions. I don't think I spent more than 5 minutes on this warm up routine, yet I sure did feel the benefit even from the first exercise onwards, where my arms were soon feeling the pump from the first set onwards.

For any of the static stretches, I wasn't pushing into the stretch hard, just enough where I could feel all the muscle fibers under tension. Then I would take a few deep, slow and controlled breathes, and release.

Beginner Bicep Summary

In summary, the beginner bicep routine is simply to familiarize yourself with several key movements that target both the long and short head of the biceps, as well as unilateral and isolation work.

After spending several minutes warming up the biceps with some light resistance work, stretching and mobilization, perform each exercise for 3 sets, aiming for 10-12 repetitions on each set, and only increasing the weight to the next increment up if you really feel you can manage more weight after comfortably achieving the final few reps. All sets should be performed as straight sets and aiming for 70-85% maximum intensity.

Perform this arm workout for 4-6 weeks, aiming to add a few extra repetitions, or increase up to the next weight increment. When you can no longer increase the weight or go beyond 12 repetitions with each exercise, you're ready to move on to the intermediate bicep routine, where I'll be introducing some additional training methods.

Workout Research – Lateral Raises

[Link to the website](#)



Dumbbell Lateral Raise Video Exercise Guide

Categories: [Exercise Videos](#) [Shoulders](#)
10.9M Views



EXERCISE PROFILE

Target Muscle Group	Shoulders
Exercise Type	Strength
Equipment Required	Dumbbell
Mechanics	Isolation
Force Type	Pull
Experience Level	Beginner
Secondary Muscles	None

TARGET MUSCLE GROUP

Shoulders



Dumbbell Lateral Raise Overview

The dumbbell lateral raise is a [shoulder exercise](#) used to strengthen the muscles of the shoulder. It's a good exercise for building width in your upper body which gives you the "V" shape.

Dumbbell Lateral Raise Instructions

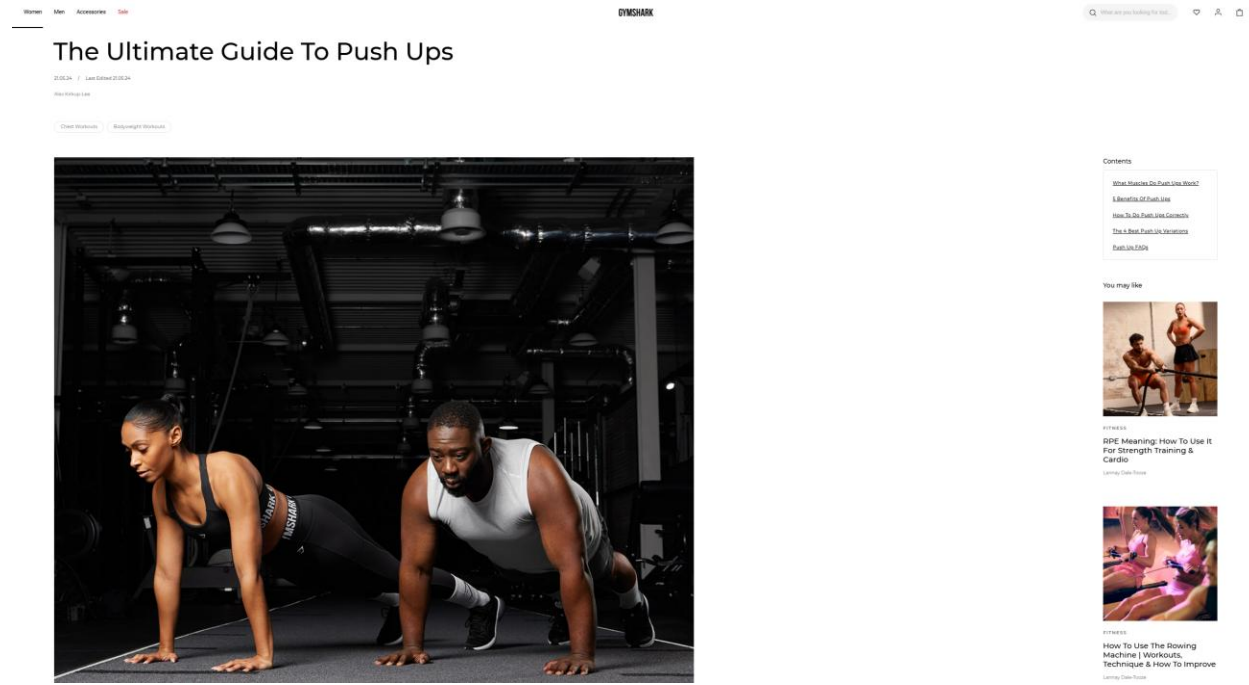
1. Select the desired weight from the rack, then take a few steps back into an open area.
2. Take a deep breath and raise the dumbbells to shoulder height using a neutral grip (palms facing in) while keeping the elbows slightly bent.
3. Slowly lower the dumbbells back to the starting position under control.
4. Repeat for the desired number of repetitions.

Dumbbell Lateral Raise Tips

- Focus on keeping the reps slow and controlled. Because this is an isolation exercise, the focus should be on correctly working the muscle - not moving as much weight as possible.
- Don't bring your arms up too high - up to shoulder height is far enough.
- If you encounter pain in the shoulder while performing the movement, consider implementing one of the following tweaks:
 - Tilt your pinkies slightly higher than your thumbs. Imagine you're pouring out a cup of water.
 - Turn your palms forward so that your thumbs are pointing away from your body. This will put the shoulder in a more externally rotated position and potentially open up the shoulder joint to allow for a bit more movement.

Workout Research – Pushups

[Link to the website](#)



Push ups: They're simple... *they're bodyweight*. But trust us, there is nothing more humbling than being able to rep out 5, 10, 15 of them.

Sadly, it's not as easy as it looks and you aren't alone if just the mere utterance of the word 'push ups' gives you flashbacks to high school gym class – but it's time to switch that thinking.

There's more to this bodyweight classic than meets the eye: yes, they're the ultimate pec builder (even giving the **bench press** a run for its money!), but mastering the push-up comes with so many health benefits too.

So what's stopping you?

Maybe you're a beginner who feels so far away from getting your first full push-up?

Or maybe you've already built up enough strength to rep out 20 unbroken, and find yourself wondering what else you could be doing to ramp up the difficulty?

Luckily, we've got the ultimate push-up form tips, plus the best variations to help you reach your first push-up, or challenge you if you're looking for something more.

So before you run off and take Drake's advice to 'Drop and Give Me 50', take a moment to read this guide. We're going to give you the steps to progress your push ups like never before – and bring back this high school gym classic, with style.

What Muscles Do Push Ups Work?

This may be a bodyweight exercise but don't underestimate the effectiveness of this compound exercise in building a strong upper body. Push-ups are one of the best chest exercises out there (especially when it comes to bodyweight training); but if you're in the gym and looking for some more exercise to hit your pecs, try these best chest exercises that should be in every chest workout.

Push ups muscles worked include:

- Chest (pecs)
- Shoulders (delts)
- Triceps

You'll probably notice that your abs get a good workout too during push ups – especially if you consciously 'squeeze' them. This will help you hold a stable plank position whilst protecting your lower back and building those six pack muscles too.

Do Push Ups Work Biceps?

Whilst the biceps help to stabilize the shoulder and elbow joints during the lowering phase of a push up, they don't directly work the biceps. If your goal is to build your biceps, you'd be better off focusing on bicep isolation exercises such as bicep curls, hammer curls, or chin ups.

5 Benefits Of Push Ups

- **Effective muscle builder:** You don't need heavy weights to build muscle: Research published in The Journal of Exercise Science & Fitness proved press-ups to be just as effective as their rival, the bench press, in increasing muscle hypertrophy and strength gains when lifted at 40% 1RM over 8 weeks [1].
- **Work multiple muscles at once:** Push-ups are a great upper body builder, but your core, legs, and glutes will get involved too. As a compound exercise, this makes them more 'functional' than isolation movements, meaning they better mimic everyday activities – making you more prepared for the challenges of everyday life!
- **Can be done from anywhere:** You've guessed this one already: Push-ups don't require anything other than your own body weight. You've got absolutely no excuses for not getting them done, wherever you are.
- **Make them easier or harder:** Push-ups are suitable for all levels, meaning beginners can build strength by elevating their hand position or dropping onto their knees, whilst more experienced gym goers can increase the intensity of push-ups by adding in plyometrics or a decline. As you become stronger, you can upgrade to a more challenging variation so you can always keep progressing.
- **Health Booster:** Push-ups aren't just good for your muscles, but your overall health too, lowering blood pressure, and cholesterol, and decreasing the risk of disease. A study of over 1000 men found that participants who could complete at least 40 push-ups had a reduced risk of cardiovascular disease [2].

How To Do Push Ups Correctly

1. Position both hands on the floor slightly wider than the shoulders. Extend your legs, coming onto your toes in a high plank position, feet hip-width apart. Squeeze your shoulder blades together, and brace your core and glutes to keep your lower back aligned. Keep your gaze on the floor in front of you.
2. Lower your chest towards the ground by bending your elbows, keeping them tight into your side. Keep your core tight and don't let your hips drop. Bring your chest as close to the floor as you can, with the nose, chest, and belly button at the same level.
3. Pause briefly, then explosively push your hands into the floor, extending your arms to come back to the high plank position. Repeat.

Note: If you find yourself struggling to do a full push-up at all, or find any of the form points listed below slipping, you may want to switch to an incline push-up.

Push Up Form Tips: Mistakes To Avoid

- **Don't let your hips sag:** This is probably the most common push up mistake which can be fixed by building core strength and initiating proper core bracing. You may need to try doing some [core strengthening exercises](#) and master doing press ups on your knees first, before taking them up to a full plank.
- **Check your hand position:** Having your hands too wide will place unnecessary strain on your shoulders. Make sure they are always under your shoulders with your fingers facing forward. Just as you would when you [squat](#), think about 'screwing your hands into the ground' promoting external rotation of the shoulders.
- **Don't let your elbows flare out:** Another common mistake that puts the shoulders at risk of injury: Make a conscious effort to keep your elbows pinned close to your side. This will protect your shoulders and maximize muscle activation where we want it – in the chest!
- **Slower is better:** This isn't a race: Perform your push ups in a slow and controlled manner to maximize muscle activation. Taking things slower will mean you fatigue less quickly so you can perform more reps and make greater strength gains. If you find regular push ups too easy, then try one of our push up progressions.
- **Don't cheat yourself:** Performing partial push up reps that only make it down halfway isn't going to help you progress. If you find yourself struggling to move through the full range of motion, scale things down and perform a modified push up where you can move through the full range of motion. You'll see a lot more progress this way, building strength to make a regular push up easier.